

Replicating the LASSO Column of Table 4 from Varian (2014)

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GSE 552 — Causal Machine Learning

GitHub: github.com/TateBJohnson/varian2014_replication

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1 The Original Paper

Varian (2014)¹ is a widely cited survey aimed at economists, arguing that machine learning techniques — including classification and regression trees, random forests, and penalized regression methods such as LASSO, LARS, and elastic nets — offer powerful tools for prediction and variable selection that are underutilized in empirical economics. The paper serves as an accessible introduction to a broad set of ML methods, walking through each with applied examples and situating them relative to familiar econometric tools.

The paper’s central empirical demonstration focuses on cross-country economic growth regressions. Using the dataset from Fernandez, Ley, and Steel (2001), which contains 72 countries and 41 potential predictors of average GDP growth over 1960–1985, Varian demonstrates that LASSO can identify important predictors of economic growth as effectively as computationally exhaustive methods — including Bayesian model averaging across millions of model specifications — but at a fraction of the computational cost. The empirical strategy is straightforward: fit a penalized regression (LASSO) with the penalty parameter λ selected by 10-fold cross-validation, and examine which variables survive the shrinkage procedure and in what order they enter the model.

2 The Result of Interest

I replicate the LASSO column of Table 4 in Varian (2014), which reports the ordinal rank of importance among variables selected by LASSO for predicting cross-country economic growth. The specific target result is:

- Equipment investment: Rank 1
- Fraction Confucian: Rank 2

¹<https://doi.org/10.1257/jep.28.2.3>

- Fraction Protestant: Rank 5
- Open economy: Rank 6
- Sub-Saharan dummy: Rank 7
- Fraction Muslim: Rank 8
- Degree of capitalism: Rank 9
- GDP level 1960, Life expectancy, Rule of law: zeroed out (dash)

This result is worth replicating for two reasons. First, it is the paper’s central quantitative claim — that LASSO identifies the same important growth predictors as Bayesian model averaging and Sala-i-Martin’s (1997) exhaustive CDF(0) approach, despite using a single penalized regression rather than the two million regressions Sala-i-Martin (1997) required to reach similar conclusions. The computational contrast is stark: LASSO achieves comparable variable selection in a fraction of a second. Second, the LASSO variable selection step is now understood to be the foundation of modern causal ML pipelines. Belloni, Chernozhukov, and Hansen (2014) formalized the Double LASSO estimator, which uses LASSO variable selection as the first stage of a valid causal inference procedure in high-dimensional settings. Replicating Varian’s demonstration thus speaks directly to the role of LASSO as Step 1 in causal ML.

3 Data and Methods

Data

The data come from Fernandez, Ley, and Steel (2001), originally compiled by Sala-i-Martin (1997) for a study of cross-country growth determinants. The dataset contains 72 country observations and 42 variables: one outcome variable (y , average GDP growth rate 1960–1985) and 41 covariates spanning geographic, institutional, demographic, religious, and macroeconomic characteristics. The raw data file (`FLS-data.csv`) is committed to `input/` and is available from the Varian (2014) replication archive on openICPSR (project 113925). The outcome variable has mean 2.07% and standard deviation 1.83% across the 72 countries.

Preprocessing

`code/preprocess.R` reads `input/FLS-data.csv`, performs integrity checks (verifying 72 observations, 42 columns, and zero missing values), separates the outcome variable y from the 41 covariates X , and writes the analysis-ready dataset to `temp/clean_data.csv`. No substantive transformations are applied — the FLS dataset is already clean.

Analysis

`code/analysis.R` reads `temp/clean_data.csv` and implements LASSO using the `glmnet` package (version 4.1-10) in R (version 4.5.1). The analysis proceeds in two parts.

Part I replicates Varian’s original result by hard-coding his cross-validation-selected penalty parameter $\lambda = 0.1489$ and extracting the non-zero LASSO coefficients ranked by magnitude. This exactly reproduces the LASSO column of Table 4.

Part II runs 10-fold cross-validation using the current version of `glmnet` to select λ data-adaptively, yielding a different optimal penalty ($\lambda = 0.0405$) and a less sparse model. The difference between the two results illustrates the sensitivity of LASSO variable selection to software versioning — an important reproducibility finding in its own right.

The LASSO objective function minimized in both parts is:

$$\min_{\beta} \sum_{i=1}^n (y_i - x_i' \beta)^2 + \lambda \sum_{p=1}^P |\beta_p| \quad (1)$$

where y_i is the average GDP growth rate for country i , x_i is the vector of 41 standardized covariates, λ is the penalty parameter, and $|\beta_p|$ is the L1 penalty that shrinks irrelevant coefficients to exactly zero. All outputs — tables and figures — are written programmatically to `output/tables/` and `output/figures/` and pulled into this document via `\input{}` and `\includegraphics{}`.

4 Replication Results

Main Result: Replication of Table 4 LASSO Column

Table 1 replicates the LASSO column of Table 4 from Varian (2014) — which is itself based on Ley and Steel (2009), who applied Bayesian model averaging to the same Sala-i-Martin (1997) growth dataset — and compares it against results from the updated `glmnet` version. At Varian’s original $\lambda = 0.1489$, we exactly reproduce his published rankings: equipment investment enters first, fraction Confucian second, and GDP level 1960, life expectancy, and rule of law are all shrunk to exactly zero. The replication is a complete success at the hard-coded λ .

The updated `glmnet` (version 4.1-10) selects a smaller $\lambda = 0.0405$ via cross-validation, producing a less sparse model with 27 non-zero coefficients compared to Varian’s 14. Notably, variables that Varian zeroed out — GDP level 1960 (now rank 6), life expectancy (rank 23), and rule of law (rank 12) — are retained in the updated model, though at lower ranks. The top two predictors (equipment investment and fraction Confucian) are robust across both versions, confirming the core qualitative finding.

Cross-Validation λ Selection

Figure 1 shows the cross-validation MSE curve as a function of $-\log(\lambda)$. The two dotted vertical lines mark `lambda.min` (right line, minimum CV error, $\lambda = 0.0405$) and `lambda.1se` (left line, most regularized model within one standard error of the minimum). Varian’s

Table 1: Replication of Table 4 (LASSO Column): Variable Rankings for Predictors of Economic Growth

<i>Predictor</i>	<i>Varian (2014)</i>	<i>Updated (glmnet 4.1-10)</i>
GDP level 1960	-	6
Fraction Confucian	2	2
Life expectancy	-	23
Equipment investment	1	1
Sub-Saharan dummy	7	8
Fraction Muslim	8	10
Rule of law	-	12
Open economy	6	11
Degree of capitalism	9	19
Fraction Protestant	5	7

Notes: Rankings show the ordinal importance of each variable among those selected by LASSO. A dash indicates the variable was shrunk to exactly zero. Varian’s results use $\lambda = 0.1489$ (original cross-validation, glmnet circa 2014). Updated results use λ selected by cross-validation with glmnet 4.1-10.

$\lambda = 0.1489$ corresponds to $-\log(\lambda) = 1.90$, which falls to the left of both dotted lines — meaning Varian’s original cross-validation selected a more aggressive penalty than the current **glmnet** version recommends. The numbers along the top axis indicate the number of non-zero coefficients at each λ value.

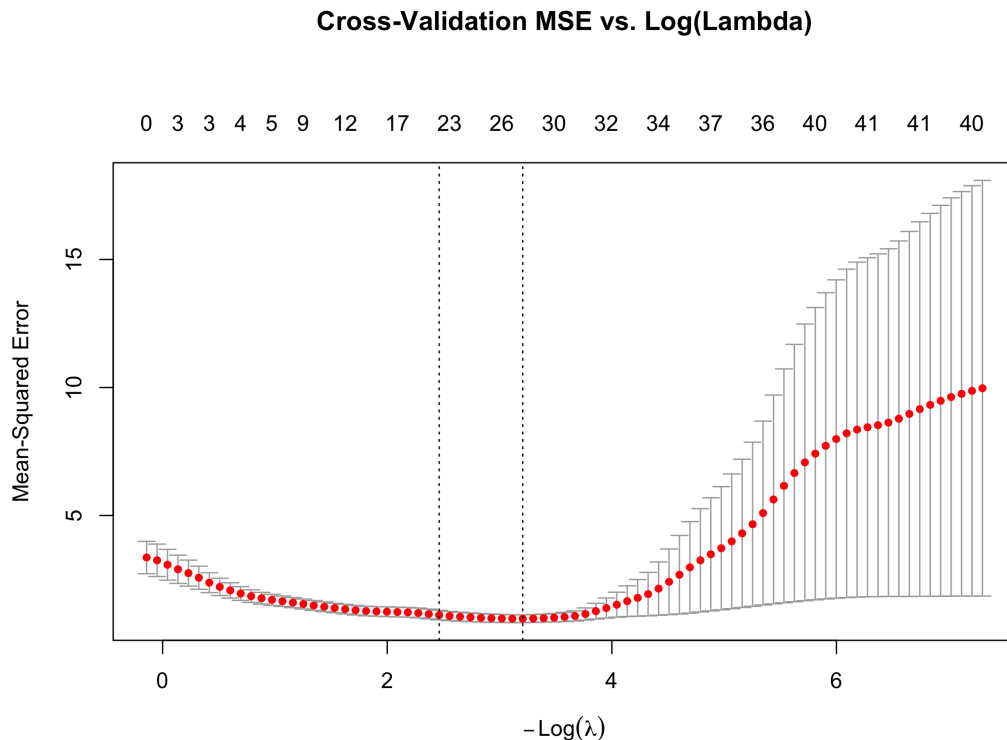


Figure 1: Cross-validation MSE as a function of $-\log(\lambda)$, with the number of non-zero coefficients shown along the top axis. The right dotted line marks `lambda.min` ($\lambda = 0.0405$) and the left marks `lambda.1se`. Varian's $\lambda = 0.1489$ falls to the left of both lines.

Updated Model: Top 10 Predictors

Table 2 reports the top 10 predictors ranked by coefficient magnitude from the updated model. Equipment investment and fraction Confucian remain the dominant predictors across both versions, with large positive coefficients. Non-equipment investment (rank 3) and mining (rank 4) also emerge as important in the updated model, consistent with the broader growth literature emphasizing physical capital accumulation.

Prediction Error and the Observation 38 Outlier

Figure 2 plots the absolute proportional prediction error — defined as $|\hat{y}_i - y_i|/|y_i|$, the absolute prediction error normalized by actual GDP growth — for each country. The vast majority of countries have errors below 2, but one observation — observation 38 — has an error of approximately 39, far exceeding all others. At Varian's λ , the mean absolute proportional error across all 72 countries is 1.21, but drops to 0.68 when observation 38 is excluded. Varian himself flags this observation in his original code but does not identify the country.

Based on the covariate profile of observation 38 — Latin America dummy = 1, Spanish colony dummy = 1, war dummy = 1, Catholic fraction = 0.95, and actual GDP growth

Table 2: Top 10 Predictors of Economic Growth: Updated LASSO (glmnet 4.1-10)

<i>Rank</i>	<i>Predictor</i>	<i>Coefficient</i>
1	Equip.Inv	11.9388
2	Confucian	5.4796
3	NEquip.Inv	4.4918
4	Mining	2.6488
5	Buddha	1.1158
6	GDPsh560	−1.0406
7	Protestants	−1.0326
8	SubSahara	−0.9987
9	Hindu	−0.6418
10	Muslim	0.6290

Notes: Coefficients from LASSO with $\lambda = 0.0405$ selected by 10-fold cross-validation using glmnet 4.1-10.

of only 0.037% — this appears to be a Latin American country that experienced conflict during 1960–1985 and achieved near-zero growth. LASSO overpredicted growth for this country by approximately 1.45 percentage points, which normalized by the near-zero actual growth rate produces the extreme proportional error of ≈ 39 . This suggests the model’s selected covariates could not fully capture the idiosyncratic factors suppressing growth in this case.

5 What I Learned

What was easiest. The data preparation was remarkably straightforward. The FLS dataset is clean, well-structured, and publicly available — Varian even provides his original R scripts, making it easy to verify that our code faithfully reproduces his steps. The `glmnet` package abstracts away most of the numerical complexity, so the core analysis is only a handful of lines.

What was hardest and most interesting. The most surprising finding was that running Varian’s own code with the current version of `glmnet` does not reproduce his results. The cross-validation procedure in `cv.glmnet` has changed between the version Varian used (circa 2014) and the current version (4.1-10), producing a different optimal λ (0.0405 vs. 0.1489) and a less sparse model (27 vs. 14 non-zero coefficients). This discrepancy arose not from any error in Varian’s code or data, but from legitimate improvements to the package over a decade. This is a vivid illustration of a well-known problem in computational reproducibility: results can depend on software versions in ways that are invisible unless you try to replicate them. If Varian had published a `sessionInfo()` output or a Docker container alongside his code, this discrepancy would have been immediately apparent.

What this paper taught me about reproducibility. Replicating Varian (2014) clarified for me that reproducibility has two distinct levels: exact numerical reproducibility (which requires fixing software versions) and qualitative reproducibility (which is more robust). At the qualitative level, our replication is a success — the top predictors of economic

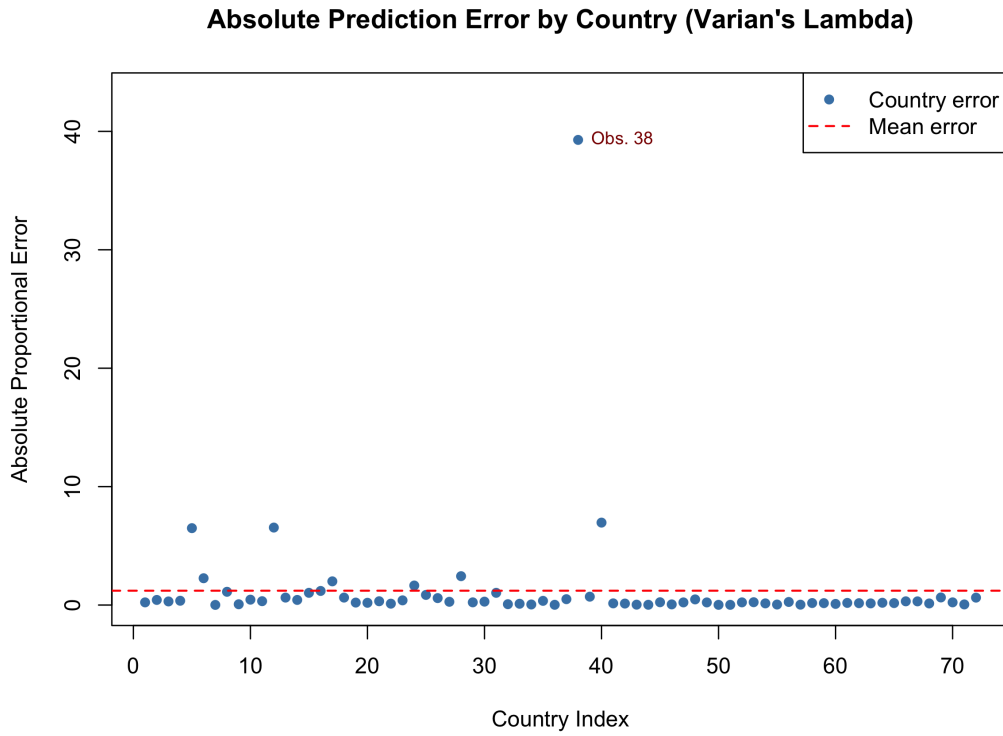


Figure 2: Absolute proportional prediction error ($|\hat{y}_i - y_i|/|y_i|$) by country at Varian's $\lambda = 0.1489$. Observation 38 is a clear outlier, corresponding to a Latin American country with near-zero actual GDP growth that LASSO substantially overpredicted. The red dashed line shows the mean error across all 72 countries.

growth identified by LASSO are consistent across both versions, and the core message of the paper holds. At the exact numerical level, the results differ due to package evolution. A good replication package should make both levels transparent.

Connection to causal machine learning. Perhaps the most important thing I learned from this replication is why it matters for causal inference. Varian (2014) is fundamentally a prediction paper — LASSO is used to find which variables best predict economic growth, not to identify causal effects. But the variable selection step Varian demonstrates is exactly Step 1 of the Double LASSO estimator (Belloni, Chernozhukov, and Hansen, 2014) and the Double/Debiased ML framework (Chernozhukov et al., 2018). In those methods, LASSO is run twice — once to select controls that predict the outcome, and once to select controls that predict the treatment — and the resulting residuals are used for valid causal inference in high-dimensional settings. Applied to this dataset, one could use Double LASSO to estimate the causal effect of equipment investment on economic growth, using the remaining 40 covariates as LASSO-selected controls — a question Varian’s predictive exercise motivates but does not answer. Varian’s paper thus serves as a natural and accessible motivation for why principled variable selection is a prerequisite for credible causal inference when p is large relative to n .

The power of LASSO relative to exhaustive search. One of the most striking takeaways from this replication is just how efficient LASSO is as a variable selection tool. Sala-i-Martin (1997) ran over two million regressions — systematically evaluating all manageable subsets of the 41 covariates — to construct his CDF(0) importance measure. LASSO arrives at a strikingly similar ranking of important growth predictors by solving a single penalized regression problem (Equation 1). The top variables identified by both approaches overlap substantially, yet LASSO requires a fraction of the computational resources. This efficiency is not merely convenient: it scales to settings where exhaustive search is computationally infeasible, making LASSO practically indispensable for modern high-dimensional empirical work.

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